

Genome-Wide Organization and Expression Profiling of the NAC Transcription Factor Family in Potato (*Solanum tuberosum* L.)

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Abstract

NAC [no apical meristem (NAM), *Arabidopsis thaliana* transcription activation factor [ATAF1/2] and cup-shaped cotyledon (CUC2)] proteins belong to one of the largest plant-specific transcription factor (TF) families and play important roles in plant development processes, response to biotic and abiotic cues and hormone signalling. Our genome-wide analysis identified 110 *StNAC* genes in potato encoding for 136 proteins, including 14 membrane-bound TFs. The physical map positions of *StNAC* genes on 12 potato chromosomes were non-random, and 40 genes were found to be distributed in 16 clusters. The *StNAC* proteins were phylogenetically clustered into 12 subgroups. Phylogenetic analysis of *StNAC*s along with their *Arabidopsis* and rice counterparts divided these proteins into 18 subgroups. Our comparative analysis has also identified 36 putative TNAC proteins, which appear to be restricted to Solanaceae family. *In silico* expression analysis, using Illumina RNA-seq transcriptome data, revealed tissue-specific, biotic, abiotic stress and hormone-responsive expression profile of *StNAC* genes. Several *StNAC* genes, including *StNAC072* and *StNAC101* that are orthologs of known stress-responsive *Arabidopsis* *RESPONSIVE TO DEHYDRATION 26 (RD26)* were identified as highly abiotic stress responsive. Quantitative real-time polymerase chain reaction analysis largely corroborated the expression profile of *StNAC* genes as revealed by the RNA-seq data. Taken together, this analysis indicates towards putative functions of several *StNAC* TFs, which will provide blue-print for their functional characterization and utilization in potato improvement.

Key words: abiotic stress; genome-wide analysis; Illumina RNA-seq; NAC transcription factor; potato

1. Introduction

Potato (*Solanum tuberosum* L.) is the most important non-grain food crop and is central to global food security. Considering its importance, much research on potato has been carried out during last decades. However, the fact remains that the global average yield of potato (15 tons/ha) is far below its yield potential (120 tons/ha), primarily due to various biotic and abiotic stresses.¹ High and low temperatures, salinity and drought are

the major abiotic stress factors limiting growth and productivity of the potato crop.^{2,3} Among biotic stresses, oomycete *Phytophthora infestans* that cause late blight is the most devastating disease of the potato with potential of causing 40–50% yield loss.⁴ Thus, improved tolerance of potato to these stresses may significantly increase the potato production. Tolerance or susceptibility against these stresses is governed by plant's ability to express a set of genes whose expression is often regulated by specific transcription factors (TFs).

The NAC [no apical meristem (NAM), *Arabidopsis thaliana* transcription activation factor [ATAF1/2] and cup shaped cotyledon (CUC2)] TFs were originally identified from consensus sequences from petunia

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NAM, *Arabidopsis thaliana* ATAF1 and 2 and CUC2.⁵ The NAC family is one of the largest plant-specific TF families, represented by 117 genes in *Arabidopsis* and 151 in rice,⁶ 163 in poplar,⁷ 152 each in soybean⁸ and tobacco⁹ and 74 in grape.¹⁰ NAC proteins regulate a variety of plant developmental processes, such as the development of shoot apical meristem,^{11,12} lateral root development,¹³ embryonic and floral development,^{12,14} stress-induced flowering,^{15,16} leaf senescence,¹⁷ regulation of cell cycle,^{18,19} hormone signalling^{13,18,20,21} and grain nutrient remobilization.²²

Some NAC proteins also regulate plant stress responses, including both biotic and abiotic.^{23,24} The *Arabidopsis* *RESPONSIVE TO DEHYDRATION 26* (*RD26*) cDNA was first identified as dehydration responsive gene²⁵ that was later shown to encode a NAC TF and functions in a novel abscisic acid (ABA)-dependent stress-signalling pathway.²⁰ Using yeast one hybrid, three *Arabidopsis* NAC proteins (ANAC019, ANAC055 and ANAC072/*RD26*) were identified, and overexpression of either of these genes significantly improved drought tolerance of transgenic plants.²⁶ Similarly, overexpression of various NAC genes in transgenic rice conferred improved tolerance against abiotic stresses.^{27–32} As far as crops are concerned, most of the studies reporting the overexpression of NAC genes are limited to rice except, by Xue *et al.*³³ who have overexpressed a wheat NAC gene, *TaNAC69* in transgenic wheat that resulted in improved dehydration tolerance. Thus, it is important to identify and functionally characterize NAC TF families from economically important crop plants and to use functional NAC genes for generating these crops with improved stress tolerance. The NAC proteins also regulate plant response against various biotic cues, including viral,³⁴ bacterial and fungal pathogens.³⁵

Typically, NAC proteins possess a conserved N-terminal DNA-binding NAC domain, which is divided into five subdomains (A–E), while C-terminal region is highly diversified and contains a transcriptional regulatory domain (TRD).³⁶ Some NAC proteins, referred as NTL (NAC with Transmembrane Motif 1-like), also contain transmembrane motifs (TMs) at their C-terminal end.^{37,38} Crystal structure of the NAC domain of *Arabidopsis* ANAC019³⁹ and rice SNAC1⁴⁰ revealed the presence of a novel TF fold consisting of a twisted anti-parallel β -sheet. Recently, a new subfamily of NAC family, called TNAC, was identified in tobacco, which seemed to be restricted to Solanaceae family.⁹ The NAC domain of TNACs lacks the LPPG and YPNG motifs that are conserved in NAC family members, whereas the conserved D/EEE motif found in other NACs is replaced by D/ExE in TNACs.⁹

The recent completion of genome sequencing of the potato by the potato genome sequencing consortium (PGSC)⁴¹ provides opportunities to identify protein families at genome-wide level, to analyse them and to

utilize the potential genes for potato improvement. Recently, Jupe *et al.*⁴² have identified 438 NB-LRR genes containing nucleotide-binding (NB) and leucine rich repeat (LRR) domain in the potato genome. Similarly in a separate report, 435 NBS-encoding R genes were identified in the potato genome.⁴³ NAC TFs have not been studied in the potato, except by Collinge and Boller,⁴⁴ who found that a potato NAC gene, *StNAC*, was rapidly and strongly induced after wounding, while under *P. infestans* infection its transcript was detected only at 48 h. However, precise function of this *StNAC* remains to be elucidated.

Given the critical roles played by NAC TFs in plants, we have identified a NAC TF family in the potato genome, provided nomenclature, performed phylogenetic analysis, mapped genes onto the 12 potato chromosomes, identified membrane-bound proteins and carried out expression analysis under various developmental stages, biotic and abiotic stresses and hormone treatments. In future, this study will provide leads to functionally characterize potato NAC TFs, to utilize them for potato improvement and also to identify and characterize NAC TFs in other *Solanum* species.

2. Materials and methods

2.1. Identification of NAC gene family in potato

All the files related to potato genome sequence data used for the identification and annotation of NAC proteins were downloaded from the PGSC data sharing site (http://www.potatogenome.net/index.php/Main_Page). The Hidden Markov Model (HMM) profile of the NAM domain (PF02365) retrieved from Pfam 26.0 (<http://Pfam.sanger.ac.uk/>) was exploited to identify the putative NAC proteins in *S. tuberosum* group Phureja DM 1-3 516 R44 (DM) protein (v3.4) database using HMM search, with an expected value (e-value) cut-off of 1.0. The sequences of all identified DM protein (DMP) models were subjected to Pfam analysis to confirm the presence of NAM domain, with an e-value cut-off of 1e–3. Keyword searches in NCBI (<http://www.ncbi.nlm.nih.gov/>), UniProt (www.uniprot.org) and PlantTFDB v2.0 (<http://planttfdb.cbi.edu.cn/>) databases were also performed to identify potato NAC proteins. *Arabidopsis thaliana* orthologs for potato NAC proteins were identified using BLASTp search against *Arabidopsis* proteins TAIR10 release (<http://www.arabidopsis.org>). Prediction of membrane-bound *StNAC* proteins was performed using the TMHMM server v. 2.0 (<http://www.cbs.dtu.dk/services/TMHMM/>).

2.2. Mapping NAC genes on chromosomes, their nomenclature and gene duplication

The position of each potato NAC gene on potato chromosomes was identified using the potato genome

browser at the PGSC site. For nomenclature, prefix 'St' for *S. tuberosum* was added followed by NAC and numbered according to its position from top to bottom on the potato chromosome 1–12. Alternatively, spliced forms were represented by Arabic numbers after '.' sign. To search for potential duplicated potato NAC genes, MCScanX software was used.⁴⁵ All 56 218 potato genes were compared against themselves using BLASTp, with criterion of tabular format (-m 8) and e-value of $<1e-5$. The resulting blast hits were incorporated along with chromosome coordinates of all protein-coding genes as an input for MCScanX and classified into segmental, tandem, proximal and dispersed duplications under default criterion.

2.3. Phylogenetic analysis and identification of conserved motifs

Multiple sequence alignment of the full-length protein sequences along with three representative Arabidopsis NAC proteins, ANAC019 (AT1G52890), ANAC055 (AT3G15500) and ANAC072/RD26 (AT4G27410),²⁶ was performed using ClustalW2 program with default parameters. Phylogenetic tree was plotted using MEGA5.05 software by the Neighbor-joining method with 1000 bootstrap replicates.⁴⁶ To study the phylogenetic relationship of potato NAC proteins along with their counterparts in Arabidopsis and rice, full-length NAC protein sequences were retrieved from TAIR10 (<http://www.arabidopsis.org>) and RGAP7 (<http://rice.plantbiology.msu.edu/>), respectively, as described.⁶ Multiple sequence alignment was performed, and unrooted tree was plotted as described above. The conserved motifs in full-length NAC proteins were identified using Multiple Expectation Maximization for Motif Elicitation (MEME) program version 4.9.0, with default parameters except the maximum number of motifs to find was set to 10.⁴⁷ To predict the secondary structure of potato NAC domain, full-length NAC sequences were aligned along with the known NAC domain structures using Promals3D web program.⁴⁸ We considered three known structures of NAC domains obtained from PDB accession number, 1UT4 (*A. thaliana*), 3SWM (*A. thaliana*) and 3ULX (*Oryza sativa*), which have most of the hits of StNAC proteins by BLAST PDB (e-value of $<1e-04$ and maximum identity of $>40\%$).

2.4. Potato RNA-seq data analysis

For expression profiling of potato NAC genes, we utilized the Illumina RNA-seq data that were previously generated by the PGSC⁴¹ and analysed by Massa *et al.*⁴⁹ The RNA-seq data of 40 libraries representing a wide range of developmental stages, abiotic and biotic stress treatments and hormone treatments were generated using Illumina Genome Analyser II

platform (Supplementary Table S1).⁴⁹ Transcript abundance is expressed as fragments per kilobase of exon model per million mapped reads (FPKM) values (Supplementary Table S2). Heat maps for only those genes were generated, which have positive FPKM values in at least one or more of the samples. For the developmental stage dataset, FPKM values were $\log_2$ transformed, before generating heat maps. For abiotic, biotic stress and hormone treatments, relative expression ratios were calculated relative to their respective controls. Heat maps were generated and hierarchical clustering done using the Institute for Genomic Research (TIGR) MeV v4.4.1 software package.⁵⁰

2.5. Plant material, in vitro culture and stress treatments

The shoot cultures of potato cv Kufri Sutlej, procured from Central Potato Research Institute, Shimla (India), were maintained under *in vitro* conditions. Potato shoots were inoculated into the Murashige and Skoog⁵¹ (MS) medium through nodal cuttings and incubated under a 16-h photoperiod ($70 \pm 5 \mu\text{mol m}^{-2} \text{s}^{-1}$ photosynthetic photon flux density) at $25 \pm 2^\circ\text{C}$ and 50–60% relative humidity. After three weeks, shoots were subjected to NaCl (100 mM), polyethylene glycol 6000 (PEG, 10%), cold (4°C), heat (42°C), ABA (100 μM) and salicylic acid (SA, 300 μM) treatments for 4 and 24 h. After stipulated time, the plantlets were harvested, frozen in liquid nitrogen and stored at -80°C until used. Shoots grown on MS basal medium at 25°C served as control. For the collection of root, stem, old leaf and young leaf samples, *in vitro* raised plantlets of potato cv. Kufri Sutlej were hardened and grown under contained conditions. Two-month-old plants were uprooted, and samples were harvested and frozen in liquid nitrogen and stored at -80°C until used.

2.6. RNA isolation and quantitative real-time PCR

Total RNA was isolated from 100 mg of frozen tissue using *iRIS* solution following the method as described.⁵² First-strand cDNA synthesis was done using RevertAidTM RNase H minus cDNA synthesis kit as per manufacturer's instructions (Fermentas Life Sciences, USA). The primers for quantitative real-time PCR (qRT-PCR) analysis were designed using the Primer3 v.0.4.0 software (<http://frodo.wi.mit.edu/>; Supplementary Table S3). Reverse primers were designed preferentially from 3'-untranslated region wherever possible, because it is generally more unique than coding sequence and closer to the reverse transcriptase (RT) start site. To check the primer specificity, amplicons obtained after PCR were sequenced using the BigDye terminator sequencing kit on an automated DNA sequencer (3730 $\times$ 1 DNA Analyser, Applied Biosystems, USA). The amplicon sequences are presented in Supplementary Table S3.

The qRT-PCR assays were performed with three biological and three technical replicates. Each reaction was performed in 20 μ l reaction mixture containing diluted cDNA sample as template and 2 $\times$ Power SYBR Green PCR master mix (Applied Biosystems), and 200 nM each of forward and reverse gene specific primers. The reactions were performed using the MX 3000P Real-Time PCR system (Stratagene) with the following programme: 95°C (90 s) [94°C (30 s), corresponding annealing temperature (30 s), 72°C (30 s)] $\times$ 40 cycles. The specificity of the amplification was also determined by dissociation curve analysis in each case. To normalize the variance in cDNA input, *elongation factor 1- α* (*ef1 α*) gene was used as the internal control as suggested earlier.⁵³ The relative expression ratio of each gene was calculated using the comparative C_t value method.⁵⁴

3. Results and discussion

3.1. Identification and nomenclature of the NAC family members in potato

To identify the putative NAC proteins in potato genome, HMM search was performed using the HMM profile of the NAM domain. This HMM search resulted in identification of 145 protein models (DMPs), which were encoded by 118 gene models (DMGs; Supplementary Table S4). Subsequently, all 145 protein sequences were subjected to Pfam analysis, with e-value cut-off of $1e-3$, which resulted in identification of 136 NAC proteins encoded by 110 genes, because nine DMPs, either with no N-terminal NAM domain or with its e-value of $>1e-3$ were excluded. A keyword search against the NCBI, UniProt and PlantTFDB databases resulted in identification of 12, 7 and 40 previously annotated potato NAC proteins sequences, respectively (Supplementary Table S5). A careful analysis confirmed the presence of these proteins in the list of 136 NAC proteins identified through HMM search in potato genome. Hence, we show that potato NAC family is comprised of 136 NAC proteins, which are encoded by 110 genes (Table 1). Thus, NAC family in the potato is also comprised of >100 genes as reported for Arabidopsis, rice, poplar, soybean, tobacco, maize and grape.^{6–10} The annotations for potato NAC proteins reported in the NCBI and UniProt databases were highly disordered and uninformative (Supplementary Table S5). Thus, a uniform nomenclature has been assigned to 136 potato NAC proteins. Potato NAC proteins are designated as StNAC followed by Arabic number 1–110 based on the position of their corresponding genes on chromosomes 1–12 and from top to bottom (Table 1). Alternatively, spliced proteins are designated by same name by adding Arabic number 1, 2 and so on

after ‘.’ sign. Similar criteria have also been adapted for the nomenclature of NAC proteins in soybean⁸ and WRKY proteins in maize.⁵⁵ Of 110 StNAC genes, 19 (~17%) undergo alternative splicing (Table 1). However, in rice, of 151 NAC genes, 15 (~10%) were reported to produce alternative spliced transcripts.⁵⁶ The higher frequency of splicing events in potato NAC family than that of rice is in agreement with the previous reports, where in potato genome 9875 genes (25.3%) have been shown to undergo alternative splicing,⁴¹ whereas in rice genome, 8772 (15.7%) genes undergo alternative splicing.⁵⁷ The higher frequency of alternative splicing events in potato NAC family indicates more functional divergence of StNACs than that of rice. The length of StNAC proteins identified in this study ranges from 56 to 901 amino acids (aa) with an average of ~312 aa. Whereas, in *Populus*, the size of NAC proteins ranges from 117 to 718 aa with an average of 342 aa.⁷ In potato, the StNAC054 (56 aa) is the smallest StNAC protein, wherein NAM domain appears to be truncated at C-terminal end (Supplementary Fig. S1). Whereas, StNAC036.1 is the largest StNAC protein (901 aa) and contains two NAM domains. However, the NAM domain at its C-terminal end (StNAC036.1C) appears to be truncated lacking subdomain A and B (Supplementary Fig. S2).

In all StNAC proteins, only NAM domain is present, except in StNAC034, where an additional tyrosine kinase domain (PF07714) is also found. To check whether any NAC protein along with kinase domain is reported from any other organism, extensive BLAST searches of the NCBI database (All GenBank, EMBL, DDBJ and PDB) were performed. Interestingly, no protein was found to have NAM and protein kinase domains together, indicating that potato StNAC034 uniquely possess an additional tyrosine kinase domain. Tyrosine protein kinase catalyses ATP-dependent phosphorylation of the tyrosine residue on target proteins and plays a central role in many signalling pathways in plants.⁵⁸ The NAC proteins have been shown to physically interact with protein kinase SnRK1 α -subunits AKIN10 and AKIN11.⁵⁹ Thus, tyrosine protein kinase domain in StNAC034 may be responsible for regulating its activity by autophosphorylation. However, experimental evidences are required to establish the precise role of tyrosine kinase domain in the regulation of StNAC034 activity.

Since, Arabidopsis is considered a model plant system for plant biology research, and many of its NAC genes have been functionally characterized, its orthologous NAC proteins to StNACs have been assigned in this study (Table 1). Interestingly, this analysis has identified StNAC072 and StNAC101 as orthologs of Arabidopsis RD26 with strong e-value support. Previously, RD26 has been shown to be involved in the ABA-dependent stress-signalling pathway.²⁰ Overexpression of rice

Table 1. List of NAC transcription factor genes in potato (*Solanum tuberosum* L.) along with their corresponding proteins, CDS and protein length, duplications and *Arabidopsis thaliana* orthologs

Gene	Protein	Protein identifier	Chromosome no.	CDS length (bp)	Protein length (aa)	Duplications	At ortholog locus	At locus description	Score (bits)	e-value
<i>StNAC001</i>	StNAC001	PGSC0003DMP400000341	chr01	741	246	Dispersed	AT5G62380.1	ANAC101, VND6	47	1.00e-05
<i>StNAC002</i>	StNAC002	PGSC0003DMP400058270	chr01	423	140	Dispersed	AT4G01520.1	ANAC067	36	0.007
<i>StNAC003</i>	StNAC003	PGSC0003DMP400069271	chr01	825	274	Dispersed	AT3G10480.1	ANAC050	71	6.00e-13
<i>StNAC004</i>	StNAC004	PGSC0003DMP400031815	chr01	1212	403	Dispersed	AT2G02450.1, AT2G02450.2	ANAC034, ANAC035	312	2.00e-85
<i>StNAC005</i>	StNAC005	PGSC0003DMP400051813	chr01	588	195	Dispersed	AT5G64530.1	ANAC104, XND1	221	4.00e-58
<i>StNAC006</i>	StNAC006	PGSC0003DMP400037231	chr02	1689	562	Dispersed	AT1G65910.1	ANAC028	427	e-119
<i>StNAC007</i>	StNAC007.1	PGSC0003DMP400015241	chr02	738	245	Dispersed	AT2G17040.1	ANAC036	231	5.00e-61
	StNAC007.2	PGSC0003DMP400015242		504	167		AT2G17040.1	ANAC036	202	7.00e-53
<i>StNAC008</i>	StNAC008	PGSC0003DMP400067304	chr02	813	270	Proximal	AT5G46590.1	ANAC096	41	0.001
<i>StNAC009</i>	StNAC009	PGSC0003DMP400041300	chr02	1029	342	Tandem	AT4G27410.2	ANAC072, RD26	39	0.006
<i>StNAC010</i>	StNAC010	PGSC0003DMP400057983	chr02	1086	361	Tandem	AT5G46590.1	ANAC096	42	5.00e-04
<i>StNAC011</i>	StNAC011	PGSC0003DMP400041296	chr02	981	326	Tandem	AT5G46590.1	ANAC096	39	0.004
<i>StNAC012</i>	StNAC012	PGSC0003DMP400041297	chr02	978	325	Tandem	AT2G46770.1	ANAC043, NST1	45	6.00e-05
<i>StNAC013</i>	StNAC013	PGSC0003DMP400058560	chr02	558	185	Tandem	AT3G10500.1	ANAC055, ATNAC3	48	4.00e-06
<i>StNAC014</i>	StNAC014	PGSC0003DMP400060071	chr02	804	267	Tandem	AT3G10500.1	ANAC055, ATNAC3	37	0.02
<i>StNAC015</i>	StNAC015	PGSC0003DMP400036603	chr02	945	314	Dispersed	AT2G43000.1	ANAC042	235	4.00e-62
<i>StNAC016</i>	StNAC016.1	PGSC0003DMP400054964	chr02	882	293	Segmental	AT4G28500.1	ANAC073	332	2.00e-91
	StNAC016.2	PGSC0003DMP400054965		561	186		AT4G28500.1	ANAC073	205	2.00e-53
<i>StNAC017</i>	StNAC017.1	PGSC0003DMP400002396	chr02	972	323	Segmental	AT5G61430.1	ANAC100, ATNAC5	365	e-101
	StNAC017.2	PGSC0003DMP400002397		774	257		AT5G61430.1	ANAC100, ATNAC5	245	2.00e-65
<i>StNAC018</i>	StNAC018.1	PGSC0003DMP400002374	chr02	1191	396	Dispersed	AT5G39820.1	ANAC094	285	3.00e-77
	StNAC018.2	PGSC0003DMP400002375		1056	351		AT1G26870.1	ANAC009	213	2.00e-55
<i>StNAC019</i>	StNAC019.1	PGSC0003DMP400022332	chr02	843	280	Dispersed	AT3G01600.1	ANAC044	277	6.00e-75
	StNAC019.2	PGSC0003DMP400022333		1182	393		AT3G01600.1	ANAC044	287	8.00e-78
<i>StNAC020</i>	StNAC020	PGSC0003DMP400023688	chr03	576	191	Singleton	AT2G24430.2	ANAC039	38	0.004
<i>StNAC021</i>	StNAC021	PGSC0003DMP400060025	chr03	699	232	Tandem	AT2G02450.1, AT2G02450.2	ANAC034, ANAC035	55	3.00e-08
<i>StNAC022</i>	StNAC022	PGSC0003DMP400061582	chr03	786	261	Tandem	AT3G10490.1/ AT3G10490.2	ANAC051/ ANAC052	74	7.00e-14
<i>StNAC023</i>	StNAC023.1	PGSC0003DMP400001112	chr03	1203	400	Segmental	AT5G24590.2	ANAC091, TIP	270	1.00e-72
	StNAC023.2	PGSC0003DMP400001113		813	270		AT5G24590.2	ANAC091, TIP	240	9.00e-64
	StNAC023.3	PGSC0003DMP400001114		1917	638		AT5G24590.2	ANAC091, TIP	270	2.00e-72
<i>StNAC024</i>	StNAC024	PGSC0003DMP400032120	chr03	849	282		AT1G69490.1	ANAC029, ATNAP	312	1.00e-85
<i>StNAC025</i>	StNAC025	PGSC0003DMP400054092	chr03	753	250	Dispersed	AT3G17730.1	ANAC057	367	e-102

Continued

Table 1. Continued

Gene	Protein	Protein identifier	Chromosome no.	CDS length (bp)	Protein length (aa)	Duplications	At ortholog locus	At locus description	Score (bits)	e-value
<i>StNAC026</i>	StNAC026	PGSC0003DMP400069047	chr03	828	275	Tandem	AT2G02450.1	ANAC034, ANAC035	71	8.00e-13
<i>StNAC027</i>	StNAC027	PGSC0003DMP400067675	chr03	699	232	Tandem	AT5G46590.1	ANAC096	62	3.00e-10
<i>StNAC028</i>	StNAC028	PGSC0003DMP400062654	chr03	819	272	Proximal	AT2G02450.1, AT2G02450.2	ANAC034, ANAC035	68	6.00e-12
<i>StNAC029</i>	StNAC029	PGSC0003DMP400067767	chr03	792	263	Proximal	AT2G02450.1, AT2G02450.2	ANAC034, ANAC035	69	4.00e-12
<i>StNAC030</i>	StNAC030.1	PGSC0003DMP400033928	chr03	540	179	Segmental	AT5G07680.1	ANAC079, ANAC080, ATNAC4	270	3.00e-73
	StNAC030.2	PGSC0003DMP400033929		999	332	Segmental	AT5G61430.1	ANAC100, ATNAC5	352	2.00e-97
<i>StNAC031</i>	StNAC031	PGSC0003DMP400062169	chr04	627	208	Dispersed	AT5G46590.1	ANAC096	50	1.00e-06
<i>StNAC032</i>	StNAC032	PGSC0003DMP400005111	chr04	852	283	Dispersed	AT1G69490.1	ANAC029, ATNAP	316	1.00e-86
<i>StNAC033</i>	StNAC033	PGSC0003DMP400055618	chr04	912	303	Tandem	AT1G01720.1	ANAC002, ATAF1	324	4.00e-89
<i>StNAC034</i>	StNAC034	PGSC0003DMP400009745	chr04	945	314	Dispersed	AT3G47570.1 / AT5G53950.1	LRR Protein Kinase / ANAC098, CUC2	92/43	3e-19 / 3e-04
<i>StNAC35</i>	StNAC35	PGSC0003DMP400058145	chr04	531	176	Dispersed	AT5G17260.1	ANAC086	43	1.00e-04
<i>StNAC036</i>	StNAC036.1	PGSC0003DMP400054265	chr04	2706	901	Segmental	AT1G34190.1	ANAC017	305	1.00e-82
	StNAC036.2	PGSC0003DMP400054267		1797	598		AT1G34190.1	ANAC017	347	1.00e-95
	StNAC036.3	PGSC0003DMP400054268		1485	494		AT1G34190.1	ANAC017	292	4.00e-79
<i>StNAC037</i>	StNAC037	PGSC0003DMP400054262	chr04	762	253	Proximal	AT5G04410.1	ANAC078, NAC2	65	4.00e-11
<i>StNAC038</i>	StNAC038	PGSC0003DMP400054263	chr04	546	181	Proximal	AT5G04410.1 / AT4G35580.1	ANAC078, NAC2, NTL9	62	2.00e-10
<i>StNAC039</i>	StNAC039	PGSC0003DMP400043482	chr04	756	251	Proximal	AT5G18270.2	ANAC087	62	5.00e-10
<i>StNAC040</i>	StNAC040	PGSC0003DMP400043483	chr04	780	259	Tandem	AT5G04410.1	ANAC078, NAC2	64	1.00e-10
<i>StNAC041</i>	StNAC041	PGSC0003DMP400043484	chr04	774	257	Tandem	AT5G04410.1	ANAC078, NAC2	61	7.00e-10
<i>StNAC042</i>	StNAC042	PGSC0003DMP400013984	chr04	819	272	Dispersed	AT1G52890.1	ANAC019	91	8.00e-19
<i>StNAC043</i>	StNAC043.1	PGSC0003DMP400048436	chr04	1128	375	Dispersed	AT2G24430.1 / AT2G24430.2	ANAC038 / ANAC039	52	1.00e-06
	StNAC043.2	PGSC0003DMP400048437		1128	375		AT2G24430.2	ANAC039	49	4.00e-06
	StNAC043.3	PGSC0003DMP400048438		978	325		AT5G07680.2	ANAC080	48	1.00e-05
<i>StNAC044</i>	StNAC044	PGSC0003DMP400017509	chr04	1068	355	Dispersed	AT1G76420.1	ANAC031, CUC3	294	5.00e-80
<i>StNAC045</i>	StNAC045	PGSC0003DMP400001544	chr05	1308	435	Tandem	AT1G25580.1	ANAC008	480	e-136
<i>StNAC046</i>	StNAC046	PGSC0003DMP400054481	chr05	1164	387	Dispersed	AT1G26870.1	ANAC009	315	4.00e-86
<i>StNAC047</i>	StNAC047	PGSC0003DMP400029528	chr05	1398	465	Dispersed	AT4G29230.1	ANAC075	424	e-119
<i>StNAC048</i>	StNAC048	PGSC0003DMP400002220	chr05	849	282	Dispersed	AT2G43000.1	ANAC042	255	3.00e-68
<i>StNAC049</i>	StNAC049	PGSC0003DMP400040416	chr05	1176	391	Proximal	AT3G10480.1	ANAC050	286	2.00e-77

<i>StNAC050</i>	StNAC050.1 StNAC050.2 StNAC050.3	PGSC0003DMP400040418 PGSC0003DMP400040419 PGSC0003DMP400040420	chr05	495 1608 1464	164 535 487	Tandem	AT5G04410.1 AT3G10500.1 AT3G10500.1	ANAC078, NAC2 ANAC053 ANAC053	244 384 300	2.00e−65 e−106 2.00e−81
<i>StNAC051</i>	StNAC051	PGSC0003DMP400044233	chr06	849	282	Dispersed	AT4G28530.1	ANAC074	266	2.00e−71
<i>StNAC052</i>	StNAC052	PGSC0003DMP400037408	chr06	447	148	Dispersed	AT1G12260.1	ANAC007, VND4	253	4.00e−68
<i>StNAC053</i>	StNAC053	PGSC0003DMP400030689	chr06	891	296		AT1G01720.1	ANAC002, ATAF1	386	e−107
<i>StNAC054</i>	StNAC054	PGSC0003DMP400003753	chr06	171	56	Segmental	AT1G65910.1/ AT3G03200.1	ANAC028/ ANAC045	67	2.00e−12
<i>StNAC055</i>	StNAC055	PGSC0003DMP400045251	chr06	990	329	Dispersed	AT1G71930.1	ANAC030, VND7	300	9.00e−82
<i>StNAC056</i>	StNAC056	PGSC0003DMP400062271	chr06	1566	521	Dispersed	AT3G15500.1	ANAC055, ATNAC3	49	6.00e−06
<i>StNAC057</i>	StNAC057	PGSC0003DMP400050122	chr06	918	305	Dispersed	AT3G18400.1	ANAC058	276	1.00e−74
<i>StNAC058</i>	StNAC058.1 StNAC058.2 StNAC058.3	PGSC0003DMP400055799 PGSC0003DMP400055800 PGSC0003DMP400055801	chr06	642 813 1011	213 270 336	Segmental Segmental	AT5G61430.1 AT5G61430.1 AT5G61430.1	ANAC100, ATNAC5 ANAC100, ATNAC5 ANAC100, ATNAC5	286 254 362	6.00e−78 4.00e−68 e−100
<i>StNAC059</i>	StNAC059	PGSC0003DMP400046923	chr06	1893	630	Segmental	AT3G49530.1	ANAC062	269	4.00e−72
<i>StNAC060</i>	StNAC060	PGSC0003DMP400058755	chr06	486	161	Dispersed	AT1G77450.1	ANAC032	84	6.00e−17
<i>StNAC061</i>	StNAC061	PGSC0003DMP400010437	chr06	1227	408	Segmental	AT5G22290.1	ANAC089	241	8.00e−64
<i>StNAC062</i>	StNAC062	PGSC0003DMP400012636	chr06	351	116	Dispersed	AT1G65910.1	ANAC028	187	2.00e−48
<i>StNAC063</i>	StNAC063	PGSC0003DMP400034966	chr06	855	284	Dispersed	AT4G28530.1	ANAC074	257	8.00e−69
<i>StNAC064</i>	StNAC064	PGSC0003DMP400032661	chr07	1470	489	Dispersed	AT3G15500.1	ANAC055, ATNAC3	174	2.00e−43
<i>StNAC065</i>	StNAC065	PGSC0003DMP400068365	chr07	549	182	Tandem	AT1G79580.2/ AT1G79580.3	ANAC033	60	6.00e−10
<i>StNAC066</i>	StNAC066	PGSC0003DMP400016573	chr07	567	188	Tandem	AT3G04060.1	ANAC046	59	3.00e−09
<i>StNAC067</i>	StNAC067	PGSC0003DMP400016578	chr07	819	272	Segmental	AT4G28500.1	ANAC073	320	1.00e−87
<i>StNAC068</i>	StNAC068	PGSC0003DMP400060971	chr07	516	171	Tandem	AT4G01540.1/ AT4G01540.2	ANAC068	60	7.00e−10
<i>StNAC069</i>	StNAC069	PGSC0003DMP400021925	chr07	864	287	Dispersed	AT5G53950.1	ANAC098, CUC2	211	3.00e−55
<i>StNAC070</i>	StNAC070	PGSC0003DMP400012529	chr07	615	204	Dispersed	AT1G77450.1	ANAC032	86	1.00e−17
<i>StNAC071</i>	StNAC071	PGSC0003DMP400033522	chr07	1020	339		AT3G15510.1	ANAC056, ATNAC2	306	2.00e−83
<i>StNAC072</i>	StNAC072.1 StNAC072.2	PGSC0003DMP400033523 PGSC0003DMP400033524	chr07	1071 486	356 161	Tandem Tandem	AT4G27410.2 AT3G15500.1	ANAC072, RD26 ANAC055, ATNAC3	363 293	e−100 4.00e−80
<i>StNAC073</i>	StNAC073	PGSC0003DMP400062002	chr07	855	284	Dispersed	AT2G43000.1	ANAC042	261	4.00e−70
<i>StNAC074</i>	StNAC074	PGSC0003DMP400038263	chr07	1050	349	Dispersed	AT1G56010.2	ANAC022, NAC1	321	5.00e−88
<i>StNAC075</i>	StNAC075	PGSC0003DMP400035655	chr08	402	133	Segmental	AT4G17980.1	ANAC071	69	6.00e−13
<i>StNAC076</i>	StNAC076	PGSC0003DMP400026135	chr08	987	328	Segmental	AT2G24430.2/ AT2G24430.1	ANAC038, ANAC039	288	5.00e−78
<i>StNAC077</i>	StNAC077.1 StNAC077.2	PGSC0003DMP400010296 PGSC0003DMP400010297	chr08	1032 1047	343 348	Segmental	AT2G46770.1 AT2G46770.1	ANAC043, NST1 ANAC043, NST1	278 300	5.00e−75 8.00e−82
<i>StNAC078</i>	StNAC078	PGSC0003DMP400051536	chr08	1047	348	Segmental	AT2G24430.2/ AT2G24430.1	ANAC038, ANAC039	284	6.00e−77

Continued

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Table 1. Continued

Gene	Protein	Protein identifier	Chromosome no.	CDS length (bp)	Protein length (aa)	Duplications	At ortholog locus	At locus description	Score (bits)	e-value
<i>StNAC079</i>	StNAC079	PGSC0003DMP400046613	chr08	576	191	Proximal	AT3G44350.2	ANAC061	35	0.024
<i>StNAC080</i>	StNAC080	PGSC0003DMP400046617	chr08	585	194	Proximal	AT5G64060.1	ANAC103	39	0.003
<i>StNAC081</i>	StNAC081.1	PGSC0003DMP400030569	chr08	1002	333	Dispersed	AT4G17980.1	ANAC071	283	1.00e-76
	StNAC081.2	PGSC0003DMP400030570		780	259		AT4G17980.1	ANAC071	275	2.00e-74
<i>StNAC082</i>	StNAC082	PGSC0003DMP400008400	chr08	897	298	Dispersed	AT1G62700.1	ANAC026, VND5	280	9.00e-76
<i>StNAC083</i>	StNAC083	PGSC0003DMP400021401	chr08	1113	370	Segmental	AT2G46770.1	ANAC043, NST1	304	6.00e-83
<i>StNAC084</i>	StNAC084	PGSC0003DMP400006960	chr09	633	210	Dispersed	AT5G09330.1	ANAC082	3.80E+01	0.005
<i>StNAC085</i>	StNAC085	PGSC0003DMP400018183	chr09	834	277	Dispersed	AT4G28530.1	ANAC074	275	3.00e-74
<i>StNAC086</i>	StNAC086.1	PGSC0003DMP400006339	chr09	468	155	Dispersed	AT2G18060.1	ANAC037, VND1	291	1.00e-79
	StNAC086.2	PGSC0003DMP400006341		1044	347		AT2G18060.1	ANAC037, VND1	415	e-116
<i>StNAC87</i>	StNAC87	PGSC0003DMP400019955	chr10	942	313	Dispersed	AT2G46770.1	ANAC043, NST1	310	9.00e-85
<i>StNAC88</i>	StNAC88	PGSC0003DMP400043440	chr10	1056	351	Segmental	AT3G15510.1	ANAC056, ATNAC2	310	8.00e-85
<i>StNAC089</i>	StNAC089	PGSC0003DMP400009699	chr10	591	196	Dispersed	AT5G64530.1	ANAC104, XND1	193	6.00e-50
<i>StNAC090</i>	StNAC090	PGSC0003DMP400019203	chr10	870	289		AT2G17040.1	ANAC036	300	6.00e-82
<i>StNAC091</i>	StNAC091	PGSC0003DMP400014381	chr10	1077	358	Tandem	AT1G01720.1	ANAC002, ATAF1	62	4.00e-10
<i>StNAC092</i>	StNAC092	PGSC0003DMP400014380	chr10	702	233	Tandem	AT5G04410.1	ANAC078, NAC2	40	0.001
<i>StNAC093</i>	StNAC093	PGSC0003DMP400014332	chr10	906	301	Proximal	AT5G04410.1	ANAC078, NAC2	42	5.00e-04
<i>StNAC094</i>	StNAC094.1	PGSC0003DMP400049938	chr11	1578	525	Tandem	AT5G64060.1	ANAC103	242	5.00e-64
	StNAC094.2	PGSC0003DMP400049939		1629	542		AT5G64060.1	ANAC103	244	1.00e-64
	StNAC094.3	PGSC0003DMP400049940		1629	542		AT5G09330.3/ AT5G09330.4	ANAC082	257	1.00e-68
<i>StNAC095</i>	StNAC095	PGSC0003DMP400054120	chr11	1203	400	Tandem	AT3G10480.2	ANAC050	290	8.00e-79
<i>StNAC096</i>	StNAC096	PGSC0003DMP400054118	chr11	1632	543	Tandem	AT5G04410.1	ANAC078, NAC2	394	e-109
<i>StNAC097</i>	StNAC097.1	PGSC0003DMP400016315	chr11	573	190	Dispersed	AT1G01720.1	ANAC002, ATAF1	190	4.00e-49
	StNAC097.2	PGSC0003DMP400016316		453	150		AT1G01720.1	ANAC002, ATAF1	271	1.00e-73
	StNAC097.3	PGSC0003DMP400016317		876	291		AT1G01720.1	ANAC002, ATAF1	387	e-108
<i>StNAC098</i>	StNAC098	PGSC0003DMP400001684	chr11	951	316	Dispersed	AT1G71930.1	ANAC030, VND7	298	3.00e-81
<i>StNAC099</i>	StNAC099.1	PGSC0003DMP400034078	chr11	1293	430	Segmental	AT2G27300.1	ANAC040, NTL8	239	2.00e-63
	StNAC099.2	PGSC0003DMP400034080		1236	411		AT2G27300.1	ANAC040, NTL8	238	6.00e-63
<i>StNAC100</i>	StNAC100	PGSC0003DMP400045708	chr11	786	261	Tandem	AT5G22380.1	ANAC090	247	6.00e-66
<i>StNAC101</i>	StNAC101	PGSC0003DMP400026903	chr12	1056	351	Tandem	AT4G27410.2	ANAC072, RD26	355	2.00e-98
<i>StNAC102</i>	StNAC102	PGSC0003DMP400017075	chr12	975	324	Dispersed	AT1G79580.3/ AT1G79580.2	ANAC033	307	5.00e-84
<i>StNAC103</i>	StNAC103	PGSC0003DMP400009522	chr12	1008	335	Dispersed	AT3G18400.1	ANAC058	281	6.00e-76
<i>StNAC104</i>	StNAC104	PGSC0003DMP400027999	chr12	474	157	Dispersed	AT3G04060.1	ANAC046	82	2.00e-16

<i>StNAC105</i>	<i>StNAC105.1</i>	PGSC0003DMP400029635	chr12	1761	586	Segmental	AT1G34190.1	ANAC017	358	5.00e-99
	<i>StNAC105.2</i>	PGSC0003DMP400029636		1440	479		AT1G34190.1	ANAC017	300	2.00e-81
<i>StNAC106</i>	<i>StNAC106</i>	PGSC0003DMP400021076		801	266		AT5G13180.1	ANAC083	255	3.00e-68
<i>StNAC107</i>	<i>StNAC107</i>	PGSC0003DMP400064998		534	177		AT3G15500.1	ANAC055, ATNAC3	52	2.00e-07
<i>StNAC108</i>	<i>StNAC108</i>	PGSC0003DMP400007702		783	260		AT5G13180.1	ANAC083	216	2.00e-56
<i>StNAC109</i>	<i>StNAC109</i>	PGSC0003DMP400065497		894	297		AT1G77450.1	ANAC032	191	5.00e-49
<i>StNAC110</i>	<i>StNAC110</i>	PGSC0003DMP400033187		753	250		AT2G43000.1	ANAC042	180	7.00e-46

OsNAC6, ortholog of Arabidopsis *RD26*, conferred dehydration and salinity stress tolerance in rice.^{28,29} Thus, functional characterization of these *RD26* orthologs will be of immense interest.

3.2. Chromosomal distribution and duplication events among *StNAC* genes

The physical map position of 105 *StNAC* genes on 12 potato chromosomes was identified. However, five *StNAC* genes could not be anchored on any of the potato chromosomes. Similarly, out of 438 NB-LRR genes, physical map position for 370 (84%) genes was predicted on potato chromosomes.⁴² The 105 members of the *StNAC* gene family are distributed non-randomly on 12 potato chromosomes (Fig. 1). Chromosomes 2 and 4 each contains the largest number of *StNAC* genes comprising 14 members (~13%), whereas chromosome 9 contains only three members (~3%; Supplementary Fig. S3). Based on the previously defined criteria,⁴² 16 clusters comprising of 40 *StNAC* genes distributed on nine potato chromosomes were identified (Fig. 1). Chromosome 2 contains the maximum number of clusters (3) comprising of nine *StNAC* genes, whereas chromosomes 1, 5, 8 and 10 each contain single cluster. Genes belonging to a family are often distributed in clusters at certain chromosomal regions. NAC family genes in rice, poplar and soybean were also found to be distributed in clusters.⁶⁻⁸

Sequencing and analysis of the potato genome revealed that it has undergone two rounds of whole-genome duplication.⁴¹ Moreover, the large size of *StNAC* gene family suggests that it has evolved through a large number of duplication events in potato. In whole potato genome, we have identified 12083 (23.47%) genes as tandem and 4253 (8.26%) genes as segmental duplicated (Supplementary Tables S6 and S7). Among *StNAC* genes, 20 were found to be segmentally duplicated, which are located on duplicated segments on chromosomes 2, 3, 4, 6, 7, 8, 10, 11 and 12 (Table 1 and Fig. 2). Maximum five *StNACs* are located in duplicated segments on each chromosomes 6 and 8, followed by three *StNACs* on chromosome 3, and two *StNACs* on chromosome 2. Duplicated segments on chromosome 4, 7, 10, 11 and 12 each contains one *StNAC*. Interestingly, all the *StNAC* gene containing chromosomal segments have a *StNAC* gene in its duplicated segment, suggesting that all the *StNAC* genes have been retained in potato after segmental duplications. Similarly, 9 NAC genes in rice⁶ and 21 NAC genes in grape¹⁰ were found to be segmentally duplicated. In addition, 27, 10 and 46 *StNACs* were also found to be tandem, proximal and dispersed duplicated, respectively (Table 1), which might have also contributed to the expansion of the *StNAC* family.

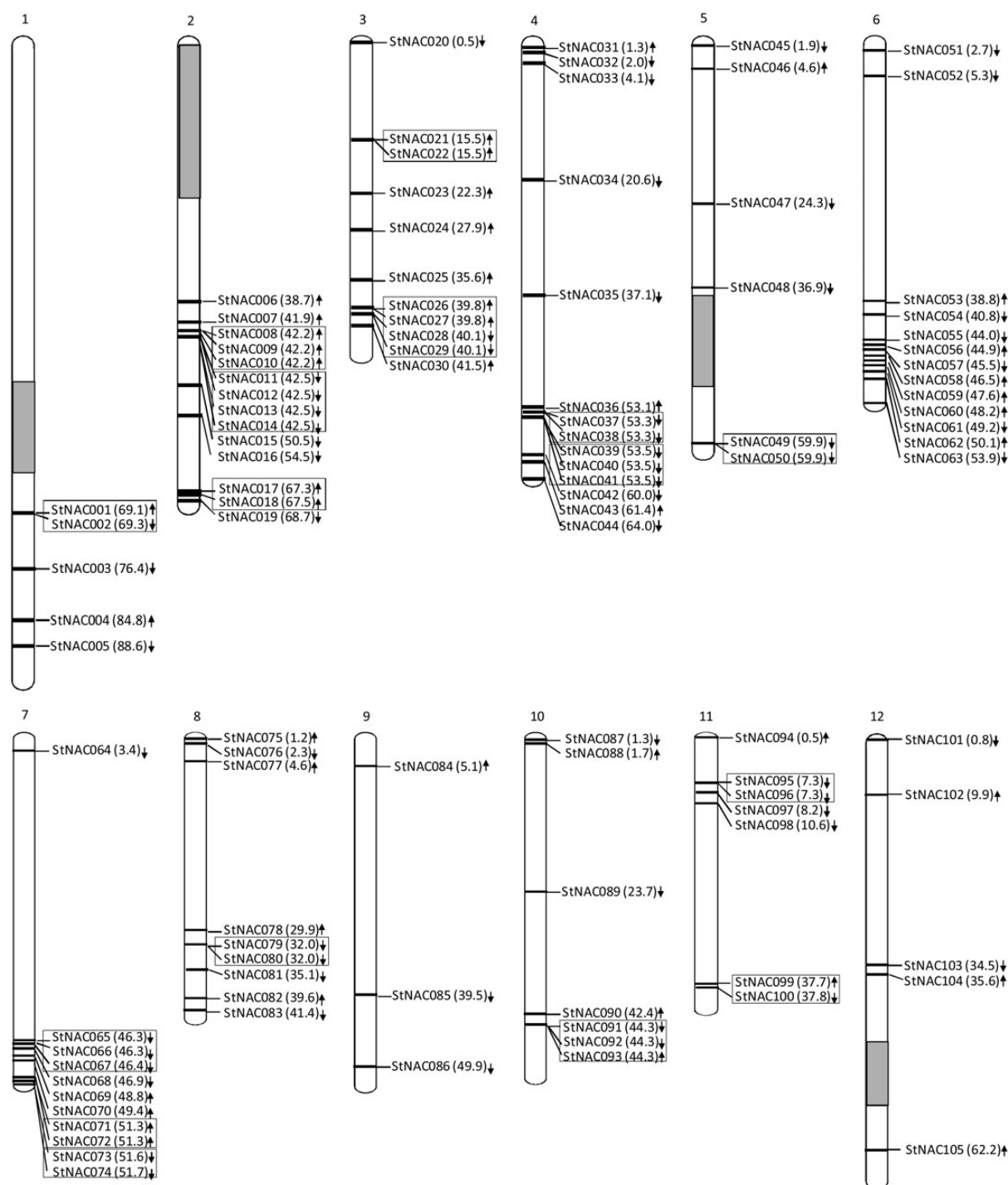


Figure 1. Chromosomal distribution of 105 potato NAC genes identified in this study. The chromosome number is indicated on the top of each chromosome. Values in parenthesis following each gene represent its position on the chromosome. Arrows pointing downward and upward represents forward and reverse orientation of the respective gene, respectively, on the chromosome. Sixteen clusters of *StNAC* genes are indicated in boxes. Grey bars on chromosome 1, 2, 5 and 12 represent known gaps in the chromosome assembly.

3.3. Structural and phylogenetic analysis of *StNAC* proteins

Multiple sequence alignment of full-length *StNAC* proteins along with three representative Arabidopsis NAC proteins,²⁶ such as ANAC019, ANAC055 and ANAC072/RD26, revealed that most of the *StNAC*

proteins contain highly conserved N-terminal NAC domain, divided into five subdomains (A– E) and a highly variable C-terminal transcriptional regulation domain as described previously (Supplementary Fig. S1).³⁶ However, of 136, 13 *StNACs* lack conserve A and/or B subdomains, and four *StNACs* do not

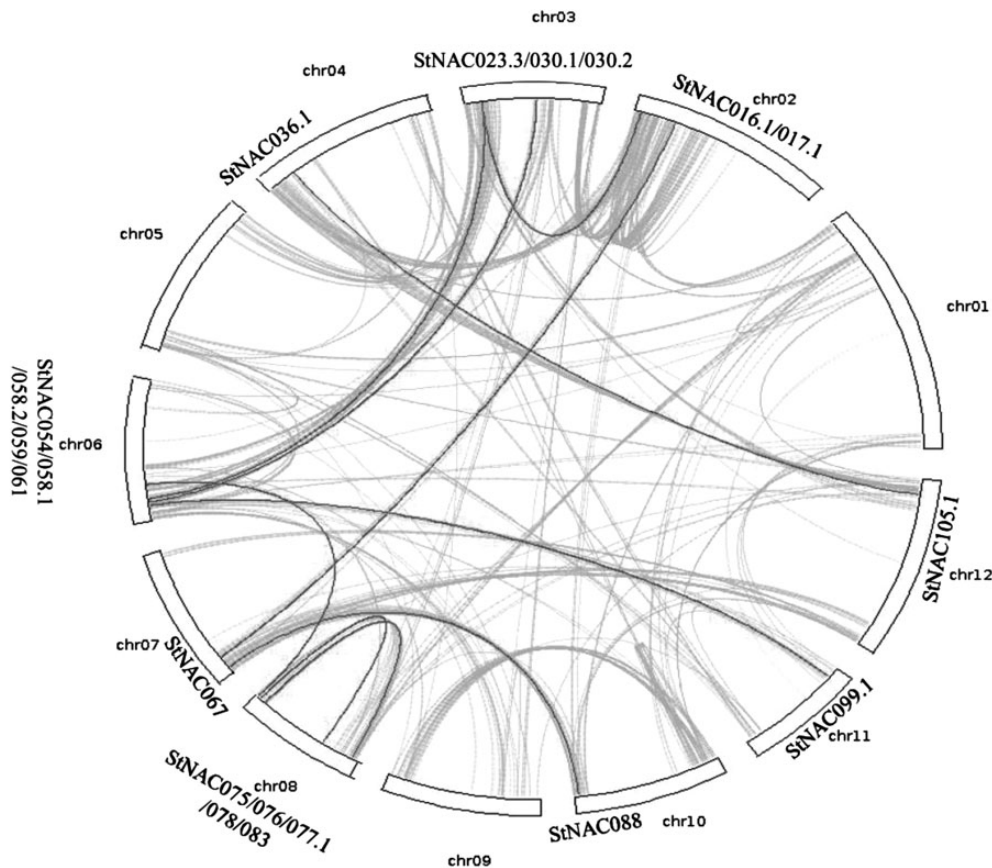


Figure 2. Depiction of segmentally duplicated StNAC genes on 12 potato chromosomes. Grey lines indicate collinear blocks in whole potato genome, and black lines indicate duplicated StNAC gene pairs.

contain conserve C and/or D subdomains. Such NAC proteins may be described as NAC-like proteins similar to the description of these proteins in soybean and rice.^{8,60} All the StNAC proteins, except StNAC054 and StNAC075, contain a conserved nuclear localization signal sequence (NLS) lying within the D subdomain. Phylogenetic tree made from multiple sequence alignment of all 136 StNAC proteins divided them into 12 distinct subgroups (Fig. 3A). Subgroup V consists of the maximum (25) number of StNAC proteins, while subgroup II, III and IV each contain minimum four StNAC proteins. In similar studies, phylogenetic analysis divided poplar and soybean NACs into 10 and 6 subgroups, respectively.^{7,8} These observations indicate that NAC proteins in potato possess more diversity than poplar and soybean. To further examine the diversity in potato NAC genes, conserved motifs were predicted by using MEME program (Fig. 3B and Supplementary Fig. S4). In general, NAC proteins clustered in same subgroups, share similar motif composition, indicating functional similarities among members of the same subgroup (Fig. 3B). Interestingly, most of the conserved motifs were found lying within the N-terminal NAC domain, indicating that these

motifs may be essential for the function of NAC proteins. While, none of the conserved motifs were found at the diversified C-terminal ends of the NAC proteins. Motifs 2, 5, 1, 3 and 6 representing the subdomains A, B, C, D and E, respectively, were present in most of the StNAC proteins. We have also predicted the secondary structure of conserved motifs corresponding to subdomains A–E covering the whole NAC domain (Supplementary Fig. S5). Previously, it was shown that NAC domain monomer consists of a twisted anti-parallel β -sheet, which packs against an N-terminal α -helix on one side and a short helix on the other side.³⁹ Similarly in our analysis, a β -sheet in subdomain B was found to be flanked with a α -helix in subdomain A and another α -helix in subdomain B. In total, six β -sheets and two α -helices were predicted, which is in agreement with the previous report.³⁹ However, in order to gain further insights into the structural features of StNAC domains, three-dimensional structure determination by X-ray crystallography would be required in future.

To examine the phylogenetic relationship of StNAC proteins with dicot (*Arabidopsis*) and monocot (rice) model plant systems, an unrooted tree was made from the alignments of full-length NAC protein

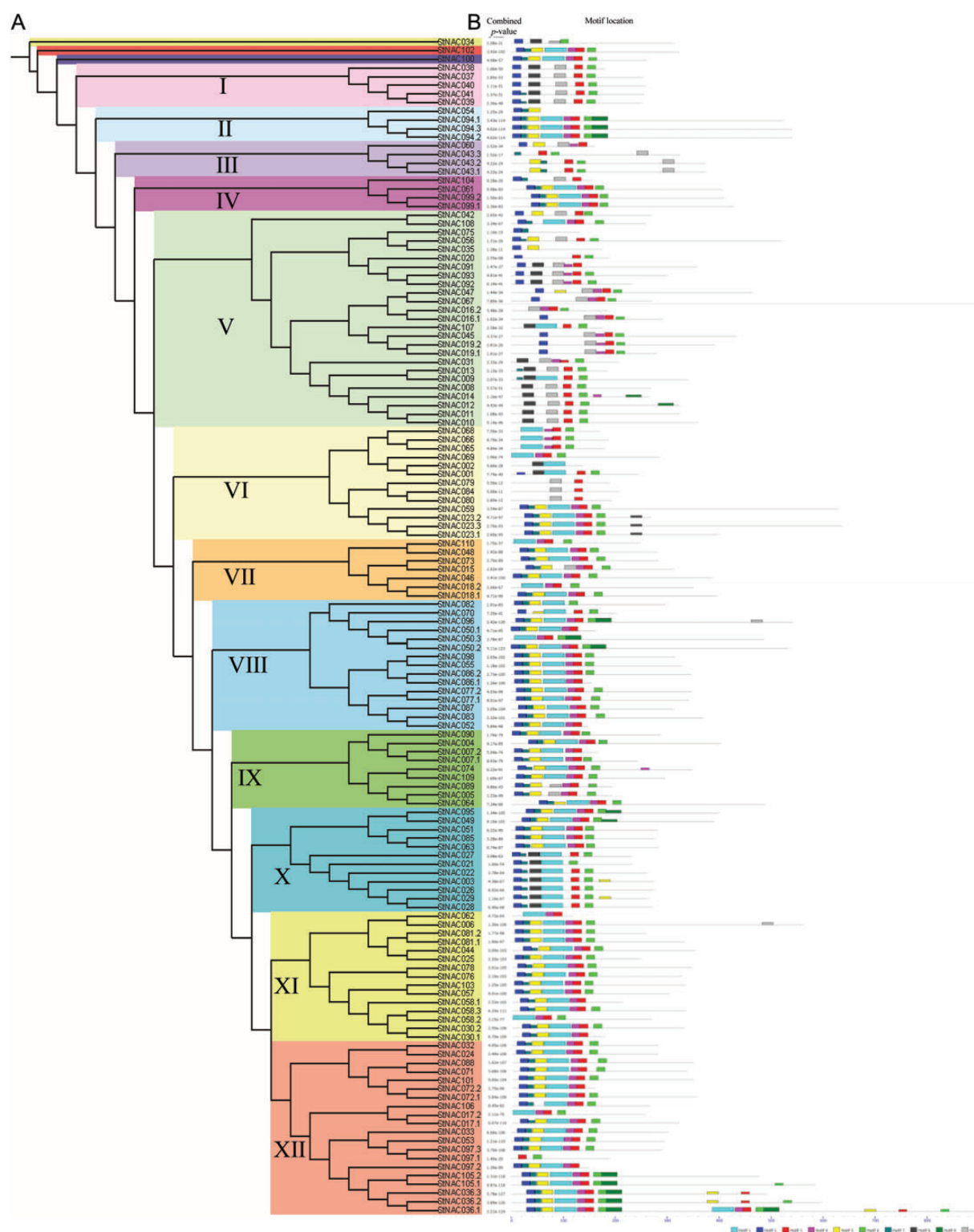


Figure 3. Phylogenetic relationship and conserved motif compositions of StNAC proteins. (A) Multiple sequence alignment of 136 full-length StNAC proteins was done using ClustalW2, and the phylogenetic tree was constructed using MEGA5.05 by the Neighbor-joining method with 1000 bootstrap replicates. The tree was divided into 12 phylogenetic subgroups designated as I to XII marked with different colour backgrounds. (B) Schematic representation of the conserved motifs in the StNAC proteins as revealed by MEME analysis. Grey lines represent the non-conserved sequences, and each motif is represented by a box numbered at the bottom. The length of protein can be estimated using the scale at the bottom.

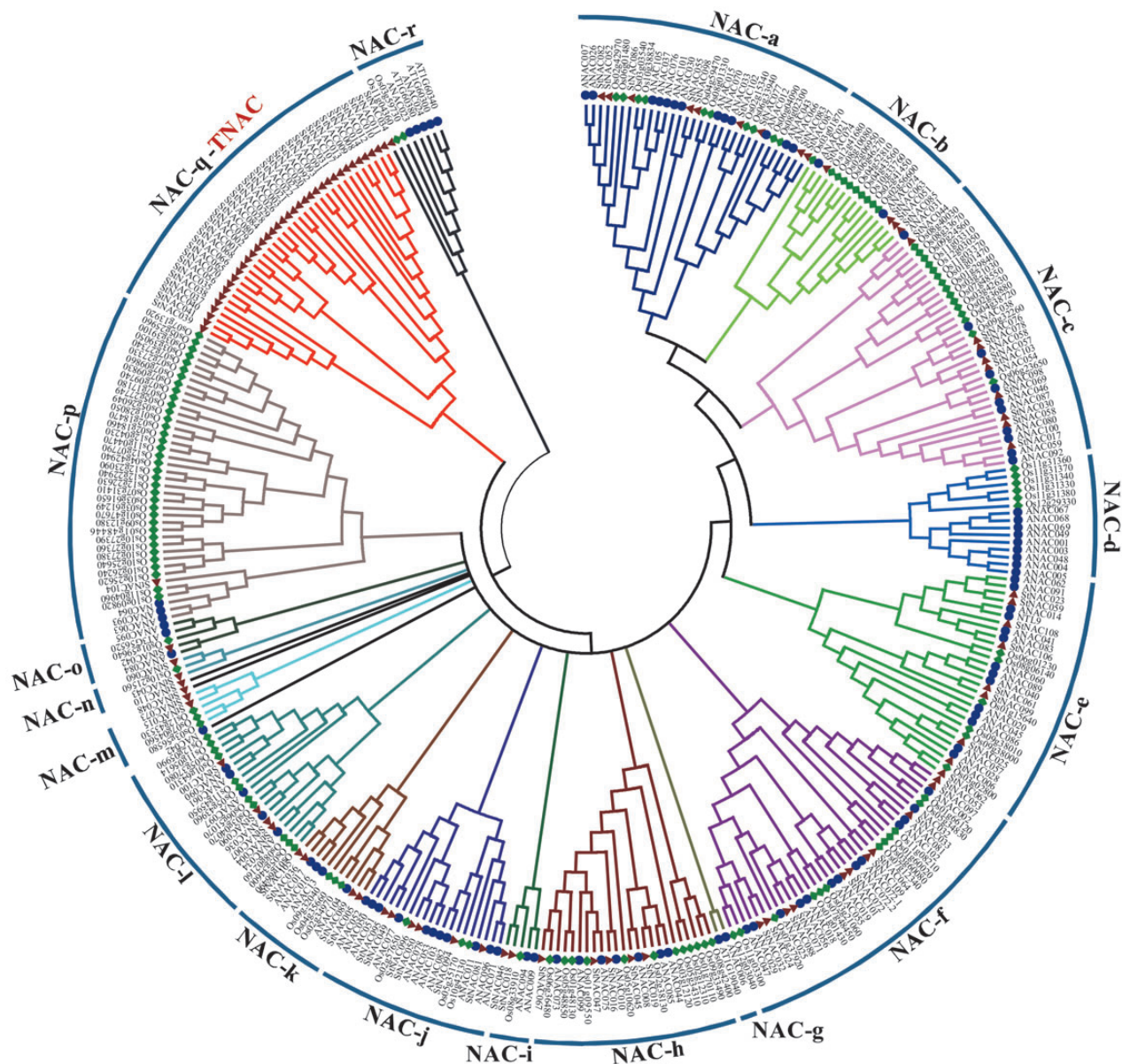


Figure 4. Phylogenetic tree of NAC proteins of potato, Arabidopsis and rice. Multiple sequence alignment of full-length NAC proteins was done using ClustalW2, and the phylogenetic tree was constructed using MEGA5.05 by the Neighbor-joining method with 1000 bootstrap replicates. The tree was divided into 18 phylogenetic subgroups, designated as NAC-a to NAC-r. Members of potato, Arabidopsis and rice were denoted by triangle, circle and diamond respectively. Subgroup NAC-q represents the TNAC subgroup, which seems restricted to Solanaceae.

sequences. The phylogenetic tree divided StNACs into 18 distinct subgroups (NAC-a to NAC-r) along with their *Arabidopsis* and rice orthologs (Fig. 4). In general, the *Arabidopsis*, rice and potato NAC proteins were distributed uniformly in all the subgroups. Exceptionally, NAC-d subgroup contains only *Arabidopsis* and rice NACs, but no potato NAC. Remarkably, NAC-q subgroup contains 36 potato NACs, but no *Arabidopsis* and rice NAC. This observation suggests that diversification and expansion of StNACs present in the NAC-q subgroup took place after the divergence of potato, *Arabidopsis* and rice. Previously, tobacco NAC family was shown to

contain a Solanaceae-specific novel subfamily, TNAC, that contains approximately 50 TNAC genes.⁹ We sought to determine whether these 36 StNACs clustered in the NAC-q subgroup belong to the TNAC subfamily. Multiple sequence alignment of NAC domain sequences of all 136 StNACs along with three representative *Arabidopsis* NACs (ANAC019, ANAC055 and ANAC072), two tobacco NACs (NCBI accession numbers BAA78417 and ADQ08688) and seven tobacco TNACs (NCBI accession numbers ACF19785, ACF19786, ACF19787, ACF19788, ACF19789, ACF19790 and ACF19791) was carried out, and an

unrooted tree was made. Interestingly, StNACs classified in the NAC-q subgroup, clustered together with tobacco TNACs (Supplementary Fig. S6), while rest of the StNACs was clustered separately along with ANACs and tobacco NACs. Thus, we suggest that these 36 StNACs may be designated as TNACs, which were also subdivided into three clades represented by A, B and C as proposed earlier.⁹ Our analysis provides further evidence that TNAC subfamily is exclusive to Solanaceae family. However, their functional characterization would be required to ascertain if they play some unique role(s) in plant processes, in which NAC proteins have not been implicated, so far.

3.4. Membrane-bound StNAC subfamily

NAC membrane-bound TFs (MTFs) have been implicated in plant response to abiotic stress.^{15,17,37} Using TMHMM server v. 2.0, we identified 14 (~10%) StNAC proteins containing α -helical TMs (Fig. 5A and Supplementary Table S8). Notably, primary transcripts of a large number of StNAC MTF genes (7 of 10) are alternatively spliced, which also code for proteins lacking the TM (Table 1 and Fig. 5A), suggesting that their activity may also be regulated at protein level through interaction between full-length and the alternatively spliced forms. Similar to Arabidopsis and rice NAC MTFs,³⁸ all the identified StNAC MTFs also contain single TM at their C-terminal (Fig. 5A). Recently in soybean, of 152 GmNACs, 11 have been predicted to contain TMs. However, GmNAC013 and GmNAC136 were found to contain two TMs.⁸ Previously, 13 members of the *Arabidopsis* NAC family were predicted to be membrane-associated and named as NTL 1–13 (for NTM1 like).⁶¹ Later, a genome-wide analysis predicted 18 NTLs in *Arabidopsis* and 5 NTLs (OsNTLs) in rice.³⁸ However, they have not assigned nomenclature for additional five *Arabidopsis* NTLs. Thus, to maintain uniformity, numbers from 14 to 18 are assigned to additional NTLs in this study. Phylogenetic analysis of the potato, *Arabidopsis* and rice NAC MTFs divided them into five clades (Fig. 5B). Maximum (14) NTLs were clustered together in Clade IV, followed by 7 each in Clades I and II, and 3 each in Clades III and V. In future, functional characterization of StNAC MTFs may identify candidate genes to engineering abiotic stress tolerance in potato and other Solanaceae plants, as well.

3.5. Differential expression of StNAC genes in various tissues/developmental stages

To identify overlapping and tissue-specific expression profile of StNAC genes, we utilized transcriptome data derived from Illumina RNA-Seq reads generated by PGSC⁴¹ and analysed by Massa *et al.*⁴⁹ The potato RNA-seq data provide the expression of over 22 000 potato

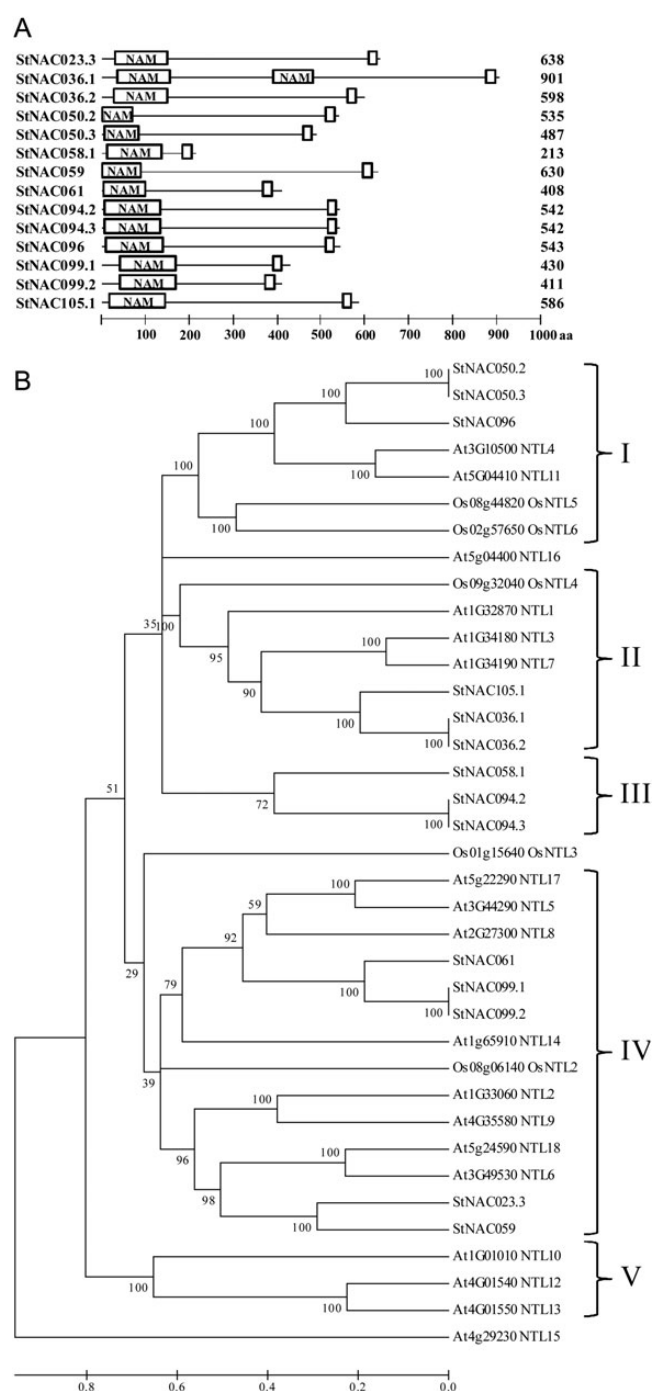


Figure 5. Membrane-bound potato NAC proteins. (A) Protein structure of membrane-bound NAC TFs. The highly conserved NAM domain is shown at the N-terminal of the proteins. α -helical TMs located at the C-terminal are shown as open box. The number of total amino acid residues in each protein is shown at the right side of each protein structure. (B) Phylogenetic relationship of membrane-bound NAC proteins of potato with that of *Arabidopsis* and rice. Multiple sequence alignment of full-length NAC MTF proteins was done using ClustalW2, and the phylogenetic tree was constructed using MEGA5.05 by the Neighbor-joining method with 1000 bootstrap replicates. The tree was divided into five phylogenetic subgroups designated as I to V. The scale at the bottom represents relative divergence of the sequences examined, and bootstrap values are displayed next to the branch.

genes in 16 tissues representing major organs and developmental stages, grouped into five major classes; floral (carpels, petals, sepals, stamens and whole mature flower), fruit (immature, mature and inside of fruit), stolon/tubers (stolons, tuber1 and tuber2), leaf (leaves, petioles) and other tissues (shoots, roots and callus).

Transcript abundance of 69 *StNACs* in 16 different developmental stages and organs was obtained, while rest of the 41 *StNACs* either transcribe at too low level to be detected or have spatial and temporal expression pattern not covered in the RNA-seq libraries. Of these 69 *StNACs*, 20 (~29%) are ubiquitously expressed in all 16 tissues, while 21 (~30%) express in 1–5, 11 (10%) in 6–10 and 17 (~15%) in 11–15 number of tissues (Fig. 6). Some of the *StNACs* also exhibit tissue-specific expression, for example, *StNAC034* and *StNAC075* express only in floral tissues, *StNAC002*, *StNAC025*, *StNAC087* and *StNAC091* in fruit tissues, *StNAC073* in stolon/tuber tissues and *StNAC082* specifically in root tissue (Fig. 6). These observations indicate that various *StNACs* may be associated with diversified functions similar to their *Arabidopsis* orthologs, for example, ANAC098 (CUC2; ortholog of *StNAC034*) regulates gynoecium development⁶² and *Arabidopsis*, vascular-related NAC domain 5 (VND5; ortholog of *StNAC082*), regulates the differentiation of root protoxylem vessels in co-operation with other VND proteins.⁶³ The tissue-specific expression profiling of *StNACs* might enable the combinatorial usage of *StNACs* in transcriptional regulation of different tissues, whereas ubiquitously expressed *StNACs* might regulate the transcription of a broad set of genes. For example, a rice NAC gene, *OsNAC10* predominantly expressed in roots and panicles and induced by drought, salinity and ABA, when overexpressed with root-specific promoter *RCc3*, improved root growth, enhanced drought tolerance and increased grain yield significantly under field drought conditions.³¹

3.6. Differential expression of *StNAC* genes during abiotic and biotic stresses

Several NAC proteins have been shown to play important roles in biotic and abiotic stress responses in plants.^{23,24} A microarray analysis in rice revealed induction of 46 NAC genes under abiotic and 26 by biotic stress.⁶ Thus, to identify the stress-responsive *StNAC* genes, we performed comprehensive expression profiling of *StNAC* genes using the Illumina RNA-Seq data. Abiotic stress treatments (24 h treatment of *in vitro* grown whole plants) include salt (150 mM NaCl), mannitol (260 μ M) and heat (35°C). Relative transcript abundance for each treatment was calculated with respect to their respective controls.

Under abiotic stress treatments, 48 *StNAC* genes express in one or more of the conditions. Of these 48 *StNACs*,

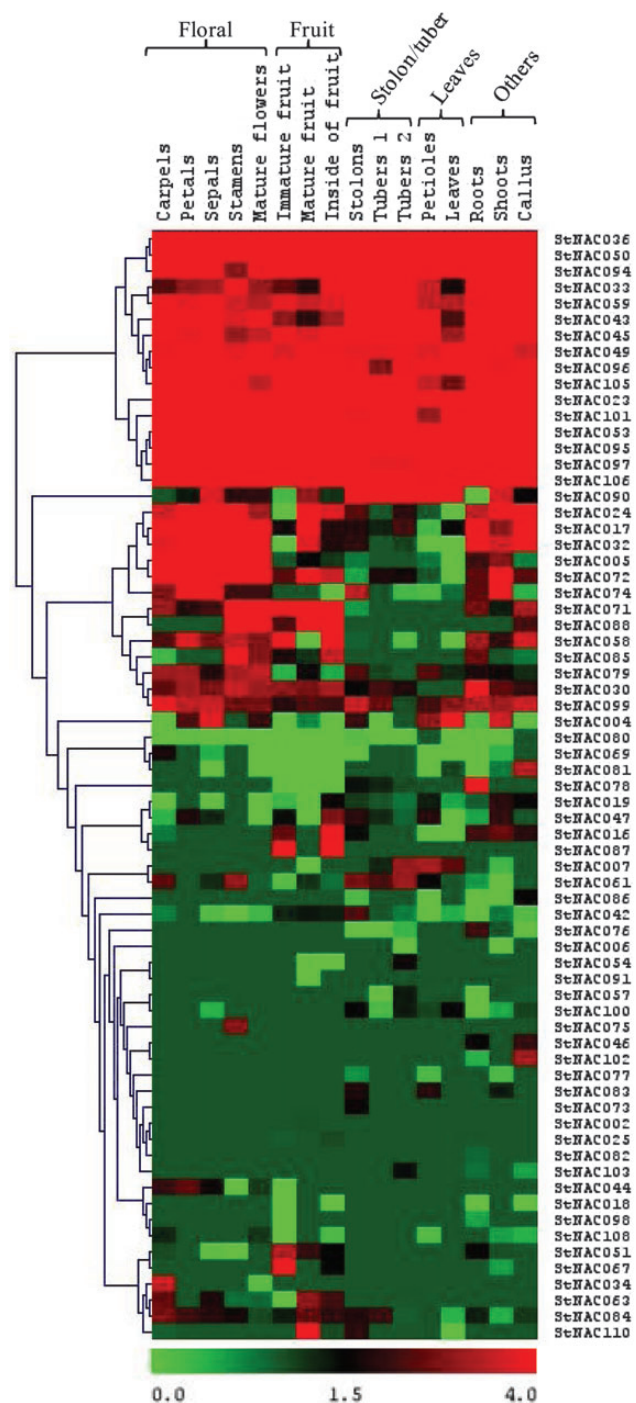


Figure 6. Heat map representation and hierarchical clustering of *StNAC* genes across different tissues and developmental stages. The Illumina RNA-seq data were reanalyzed, and the FPKM values were $\log_2$ transformed and heat map generated using TIGR MeV v4.1.1 software. Bar at the bottom represents $\log_2$ transformed values, thereby values 0, 1.5 and 4.0 represent low, intermediate and high expression, respectively.

StNAC017, *StNAC030*, *StNAC086* and *StNAC097* were found to be induced under all the three stresses, namely salt, mannitol and heat treatments (Fig. 7A). Previously,

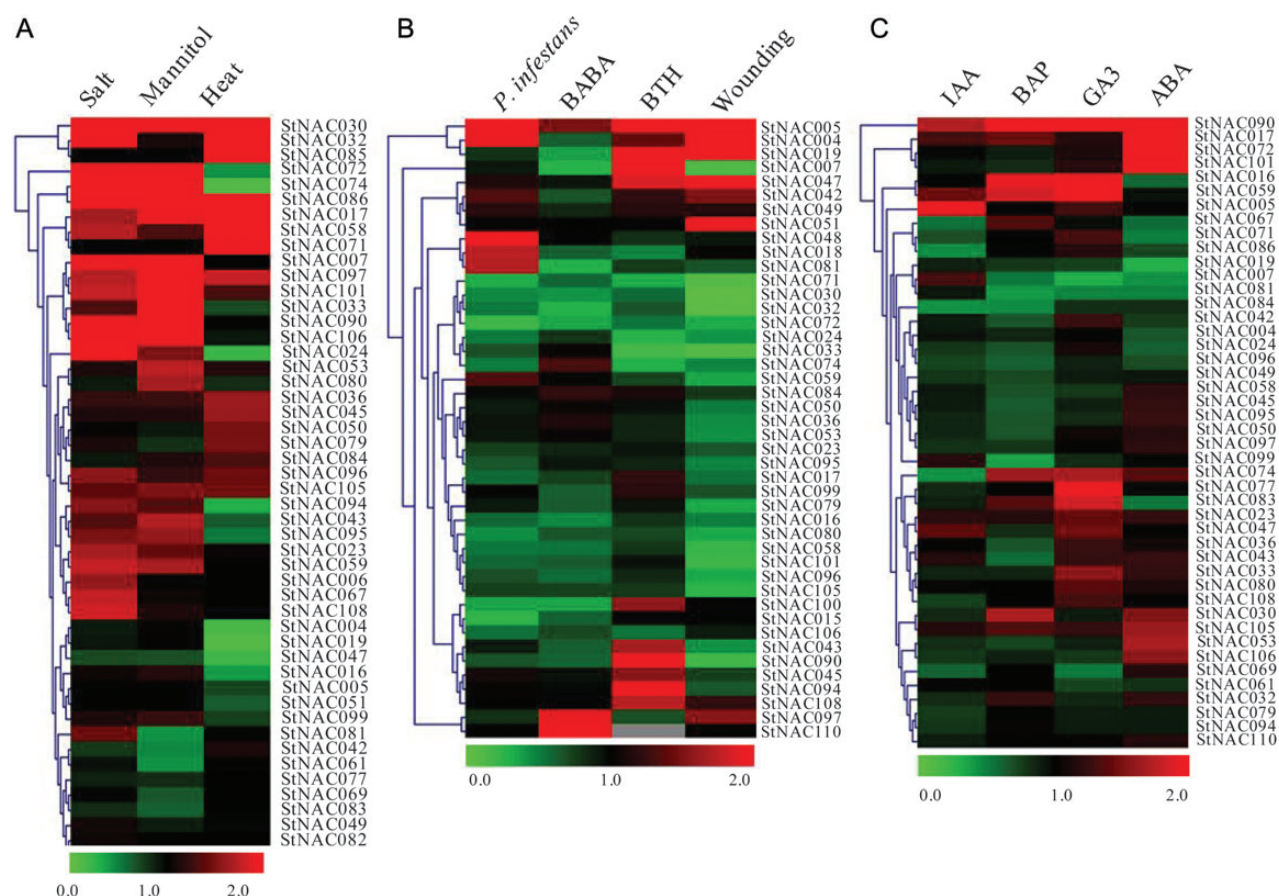


Figure 7. Heat map representation and hierarchical clustering of *StNAC* genes during (A) abiotic stress, (B) biotic stress and (C) hormone treatments. The Illumina RNA-seq data were reanalyzed, and the relative expression was calculated with respect to respective control (untreated) samples. Heat maps were generated using the TIGR MeV v4.1.1 software. Bar at the bottom of each heat map represents relative expression values, thereby values 0, 1.0 and 2.0 represent downregulated, unaltered and upregulated expression, respectively.

overexpression of multiple stress-responsive NAC genes, such as *OsNAC6*, *ONAC063*, *ONAC045* and *SNAC2*, conferred multiple abiotic stresses in transgenic plants.^{28–30} Some of the *StNACs* also exhibit induction under specific stress conditions, for example, *StNAC024*, *StNAC067* and *StNAC108* were induced specifically under salt stress, while *StNAC053* and *StNAC080* induced only under mannitol treatment and *StNAC071* and *StNAC085* induced under heat stress only (Fig. 7A). Interestingly, expression of Arabidopsis *RD26* orthologs, *StNAC072* and *StNAC101*, was highly induced by salt, mannitol and ABA treatments (Fig. 7A and C). Previously, expression of *RD26* was found to be induced by dehydration and ABA and its overexpression conferred hypersensitivity to ABA in transgenic Arabidopsis, while *RD26* repressed plants were insensitive.²⁰ Overexpression of multiple stress-responsive rice NAC gene, *OsNAC6* having high sequence similarity with Arabidopsis *RD26*, conferred dehydration and salinity stress tolerance in rice.^{28,29} Functional characterization of *RD26* orthologs identified in this study may

provide opportunities to develop abiotic stress tolerant transgenic potato and other Solanaceae crops.

The biotic stress treatments (pooled samples at 24 h, 36 h, 72 h) include induction with *P. infestans* inoculum (Pi isolate US8:Pi02–007) and two chemical elicitors, acibenzolar-s-methyl (BTH, 100 µg/ml) and DL-β-amino-*n*-butyric acid (BABA, 2 mg/ml), using detached leaves and wounded leaves to mimic herbivory. A total of 44 *StNACs* were found to be expressed in one or more of the biotic stress conditions (Fig. 7B). Interestingly, *StNAC005* was found to be induced under all the biotic stress conditions, except BABA treatment. Previously, its Arabidopsis ortholog, *ANAC104* (AT5G64530.1; Table 1) was shown to be highly induced in Arabidopsis, challenged with plant pathogen *Pseudomonas syringae* pv. *tomato* DC3000 and human pathogen *Escherichia coli* O157:H7.⁶⁴ *StNAC004* was also induced under *P. infestans* infection and wounding, but downregulated under BABA treatment. Expression of *StNAC018*, *StNAC048* and *StNAC081* was induced only under *P. infestans* infection (Fig. 7B). Expression of

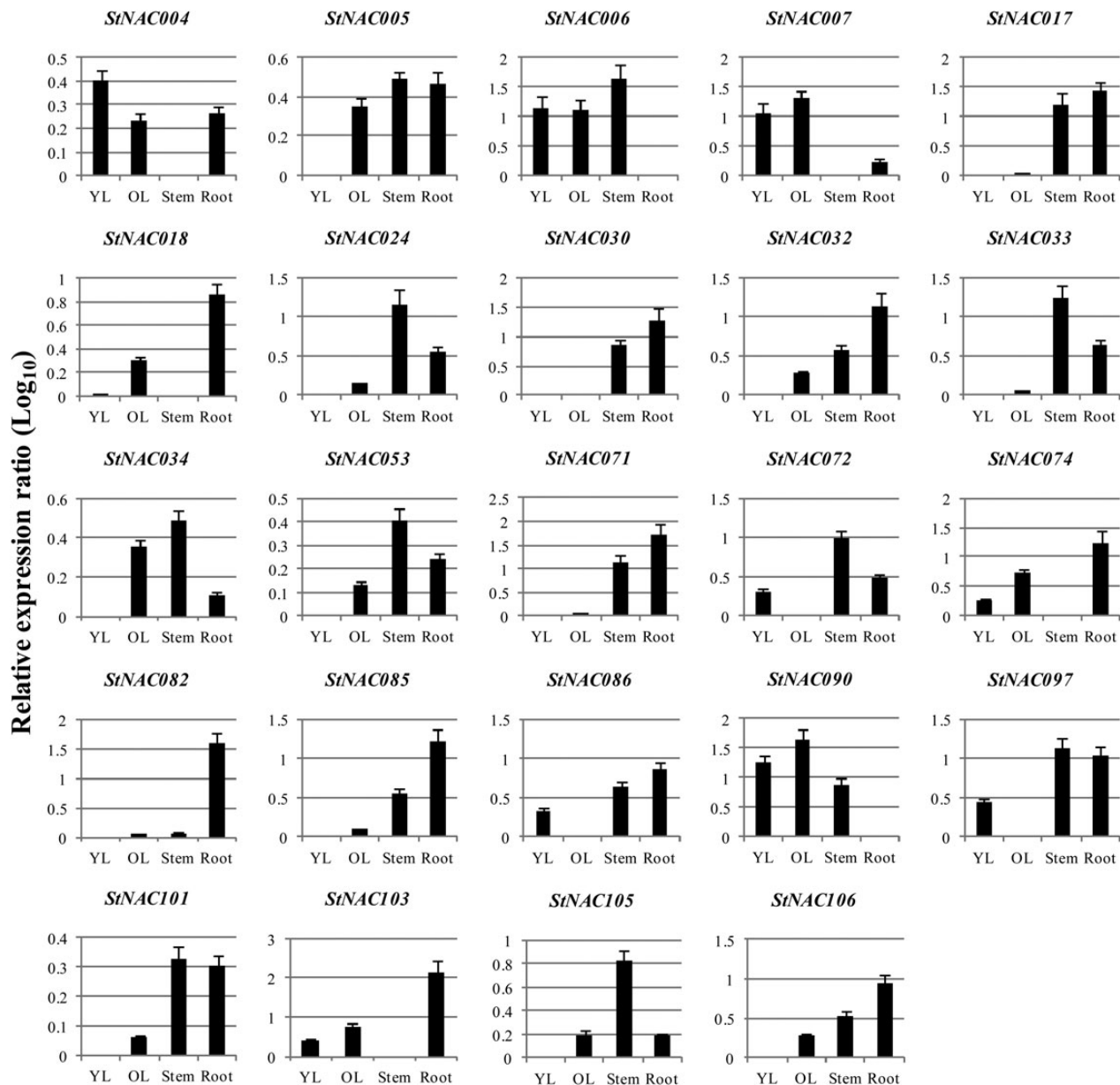


Figure 8. The relative expression ratio of 24 representative *StNAC* genes in young leaf (YL), old leaf (OL), stem and root tissues of potato determined using qRT-PCR. Relative expression ratios in different tissue samples have been calculated with reference to tissue sample in which the respective transcript exhibited the lowest expression. The relative expression values were log₁₀ transformed. qRT-PCR data were normalized using potato elongation factor 1- α gene. The name of the gene is written on the top of each bar diagram. (Error bars indicate standard deviation.)

StNAC097 and *StNAC110* was induced only under BABA treatment, whereas expression of *StNAC007*, *StNAC090* and *StNAC094* was induced only under BTH treatment. *StNAC051* was induced only under wounding stress. Previously, NAC proteins were shown to positively regulate defence response by activating pathogenesis-related genes, which in turn induce hypersensitive response and cell death at the site of infection.²¹ In contrast, NAC proteins have also been shown to negatively regulate defence response by suppressing defence-related gene expression.³⁵ In future, it would be interesting to functionally characterize these biotic

stress-responsive *StNAC* genes and to classify them as positive and negative regulators of pathogen defence response, especially against *P. infestans* infection.

3.7. Differential expression of *StNAC* genes during hormone treatments

NAC proteins have been shown to regulate a variety of plant processes by mediating hormone signalling. Thus, to identify hormone-responsive *StNAC* genes, we analysed the Illumina RNA-seq data, which include indole-3-acetic acid (IAA, 10 μ M), 6-benzylaminopurine

(BAP, 10 μ M), gibberellic acid (GA_3 , 50 μ M) and ABA (50 μ M) treatment to *in vitro* grown whole plants for 24 h.⁴⁹ Of 110 *StNAC* genes, 45 express under one or more of the hormone treatments (Fig. 7C). Interestingly, expression of *StNAC090* was induced under all the phytohormone treatments that were analysed in this study. Expression of *StNAC016* and *StNAC059* was induced under both, BAP and GA_3 treatments. In Fig. 5, we showed that *StNAC059* is a membrane-bound NAC TF. A membrane-bound, cytokinin-inducible *Arabidopsis* NAC TF, NTM1 regulates cytokinin signalling during cell division.¹⁸ Similarly, *Arabidopsis* NTL8 regulates salt-responsive flowering via *FLOWERING LOCUS T*¹⁵ and mediates salt regulation of seed germination via the GA pathway.³⁷ NTL8 expression was found to be induced by high salinity, but was unaffected by ABA. Similarly, *StNAC059* expression was induced by salt stress (Fig. 7A), but remained unaffected by ABA treatment (Fig. 7C). Interestingly, *StNAC059* and *Arabidopsis* NTL8 clustered together in Clade IV

(Fig. 5B), indicating that they also share sequence similarity with each other. *StNAC005* was induced only under IAA treatment. Overexpression of its *Arabidopsis* ortholog, *ANAC104/XND1* (AT5G64530), resulted in extreme dwarfism associated with the absence of xylem vessels and little or no expression of tracheary element marker genes. Previously, differentiation of tracheary elements was shown to be enhanced by auxin.⁶⁵ In addition, *StNAC017*, *StNAC072*, *StNAC090* and *StNAC101* were found to be highly responsive to ABA. These observations indicate that function of some of the NAC proteins might be conserved among species.

3.8. Validation of expression pattern of *StNAC* genes using qRT-PCR

Expression profiling of members of large gene families using publicly available data (for, e.g. EST, microarray, MPSS and RNA-seq data), followed by validation of the expression pattern of selected genes using qRT-

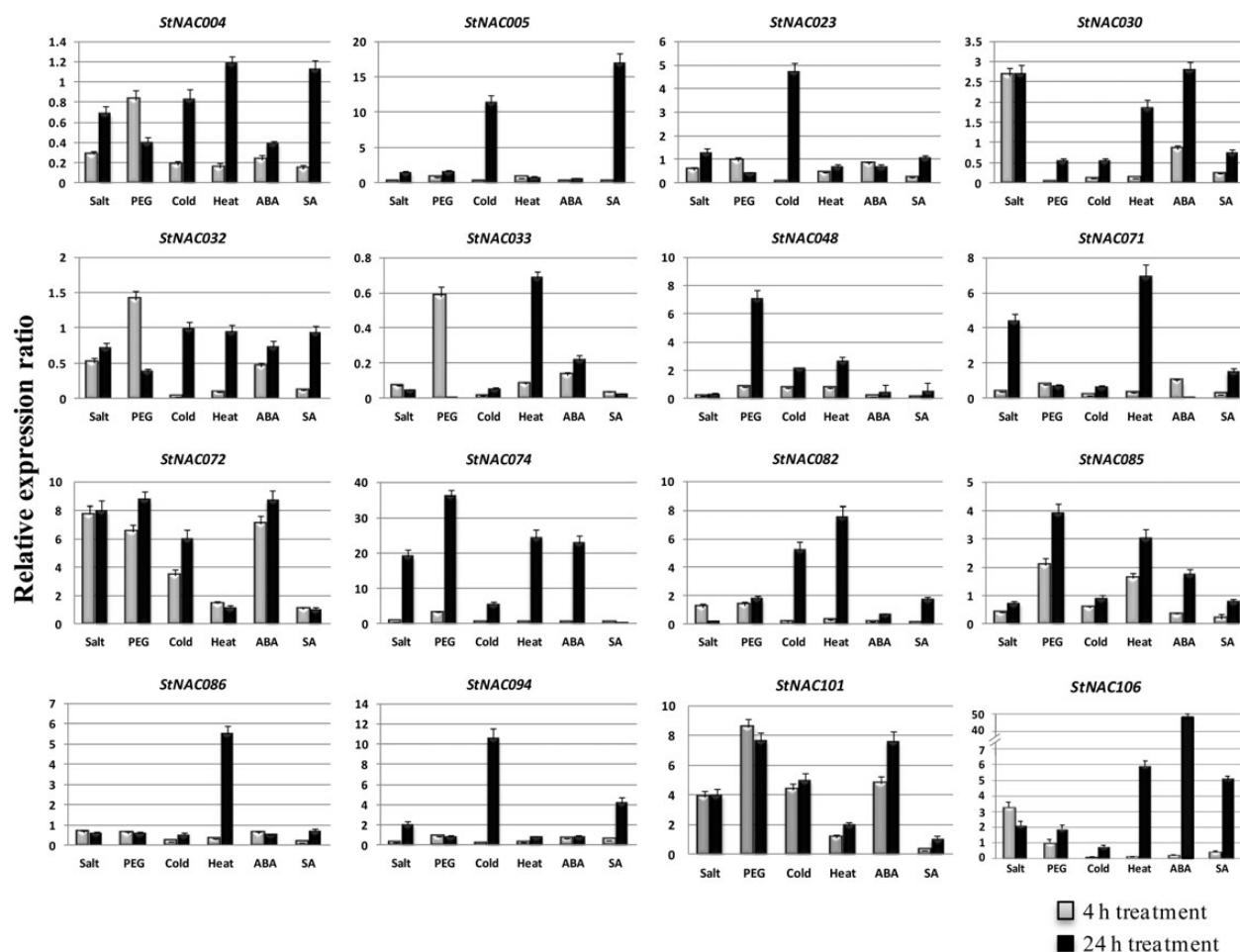


Figure 9. The relative expression ratio of 16 representative *StNAC* genes analysed by qRT-PCR under stress treatments for 4 h (grey bars) and 24 h (black bars). The relative expression ratio of each gene was calculated relative to its expression in control sample. qRT-PCR data were normalized using potato elongation factor 1- α gene. The name of the gene is written on the top of each bar diagram. (Error bars indicate standard deviation.)

PCR, is a valuable approach, which provides preliminary indications about the function of newly identified genes and often been recently exploited.^{7,8} However, in some instances, data obtained from different methods may differ. Thus, in order to validate the expression pattern of *StNAC* genes, we have carefully selected few representative *StNAC* genes with diverse expression patterns and performed qRT-PCR analysis. As shown in Fig. 8, the qRT-PCR results of (22 of 24) representative *StNAC* genes in young leaf (YL), old leaf (OL), stem and root tissues of potato were found to be largely in good agreement with the RNA-seq data (Fig. 6). However, only in case of two genes (*StNAC074* and *StNAC034*), qRT-PCR data differed from the RNA-seq data. These minor differences could be either due to difference in the stage of the plant at which the samples were collected or could be genotype dependent. For example, all the samples for RNA-seq analysis were collected from greenhouse grown plants, except root and shoot tissues, which were collected from *in vitro* grown plants,⁴⁹ whereas, in the present study, all the samples were collected from *in vitro* raised hardened plantlets grown for 2 months in greenhouse.

In another experiment, we have carried out qRT-PCR analysis of 16 representative *StNAC* genes under salt (100 mM NaCl), PEG 6000 (10%), heat (42°C) and ABA (100 µM) treatments to validate the expression pattern as revealed by RNA-Seq analysis. In addition, cold (4°C) and SA (300 µM) treatments were also included as one of the most prominent abiotic stresses and elicitor of the biotic stress response, respectively. The qRT-PCR results under these treatments also corroborate the expression profile as revealed by RNA-seq analysis. For example, expression of *StNAC030* was induced after 4 h of salt stress imposition and maintained upto 24 h, whereas its expression was induced after 24 h of heat and ABA treatment (Fig. 9), corroborating the RNA-seq data (Fig. 7). Expression of Arabidopsis *RD26* orthologs, *StNAC072* and *StNAC101*, was also found to be highly induced by stress and ABA treatments, which is in agreement with the RNA-seq data (Fig. 7A and C) and previous reports.²⁰ These results strongly suggest that preliminary expression profiling using publicly available expression data followed by its validation using qRT-PCR provide more reliable expression profile of members of large gene families in less time with reduced expenditure.

4. Conclusions

The present effort to identify and describe key attributes of uncharacterized NAC TFs in potato genome using high-throughput genome-wide survey, and utilization of available expression data coupled with molecular tools provides foundation of our

understanding of their regulatory roles. Our comprehensive genome-wide analysis led to identification of 136 NAC TF proteins encoded by 110 genes in potato. A uniform nomenclature and annotation was provided to the identified genes and proteins, followed by their comparative phylogenetic analysis with Arabidopsis and rice NAC TFs. Phylogenetic analysis led to identification of TNAC subfamily comprising of 36 *StNAC*s. Similar to tobacco, the presence of TNAC subfamily in potato provides further evidence of its existence in Solanaceae plants only. Considering the fact that most of the biological functions played by NAC TFs have been revealed using Arabidopsis NAC genes, we assigned Arabidopsis orthologs to each *StNAC* protein. The comparative analysis of *StNAC*s with their respective Arabidopsis ortholog helped us to predict the potential functions of several *StNAC* proteins. The availability of potato transcriptome data generated by the Illumina RNA-seq approach has been exploited as a useful tool for preliminary analysis of gene expression and identified tissue-specific, stress- and hormone-responsive *StNAC* genes. Additional experiments through their over- and/or under-expression will help in determining the precise function of these genes. It will also be intriguing to identify and functionally to characterize their promoters, which may be utilized to engineer potato plants with improved performance under stressful conditions, in future. Thus, this analysis provides preliminary indications of putative function of several *StNAC* genes, which will help in channelizing directional efforts for their functional characterization.

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Supplementary data: Supplementary Data are available at www.dnaresearch.oxfordjournals.org.

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